

Gabriel M. Perry

667 Wymount Terrace, Provo, UT, 84604-2044

perrygabriel99@gmail.com, gmp99@byu.edu, gperry@arlis.umd.edu

530-531-7948

<https://www.linkedin.com/in/gabriel-m-perry/>

Education

- Combined Master's/Ph.D. | Computer Science | Brigham Young University, Provo, UT | Currently Enrolled. GPA 4.00
- Bachelor of Science | Computer Science (Data Science Emphasis, minor in Mathematics) | Brigham Young University, Provo, UT | Graduated April 2024, Dean's List | GPA 3.95 (Cum Laude)
- High School (Gridley High School) | Gridley, CA | June 2017 | Valedictorian (4.17) | Eagle Scout Award

Research Experience

- Research Assistant | Information and Decision Algorithms (IDeA) Labs, Brigham Young University | 2023–Present
- 2025 RISC-ARLIS Summer Innovation Sprints | Team NY, "GeoINT Methods For Supply Chain Anomaly Detection." | Using geospatial imagery data, machine learning, and data science
- Researched for SBIR Grant | U.S. Department of Energy, "Attack Surface Modeling for Situational Awareness and Vulnerability Analysis of Power Systems" (SBIR Phase I award, contract DE-SC0021693, awarded to Achilles Heel Technologies, INC, 2021), accessed December 23, 2025, www.sbir.gov
 - Robust Control of Cyberphysical Systems | Collaborating with Provo City Power (UT), Oak Ridge National Laboratory, and KTH (Sweden) to address vulnerabilities in critical system architectures (e.g., power plants).

Publications

- Perry, Gabriel M., Micah H. Olson, David Grimsman, and Sean Warnick. "A Matrix Inverse Algorithm based on Bi-Directional Graph Search." In 2024 IEEE 63rd Conference on Decision and Control (CDC), pp. 8926-8931. IEEE, 2024.
- Sundwall, Tanner, Gabriel M. Perry, David Grimsman, and Jonathan Sillito. "A bilevel approach to resource allocation for utility-based request-response systems." In 2025 American Control Conference (ACC), pp. 2268-2273. IEEE, 2025.

Technical Skills

- Programming | Python (graduate-level mastery), C++, Java, MATLAB (undergraduate-level mastery).
- Machine Learning | Experience with PyTorch, TensorFlow, and autograd tools for deterministic and data-driven modeling and parameter learning.
- Mathematics | Robust background in control theory, game theory, cryptography, linear algebra, and optimization.
- Systems Knowledge | Coursework in computation theory (Turing machines, NP-completeness), discrete structures (boolean logic, relational algebra), and systems programming (Unix command line, memory abstraction).

Teaching and Mentorship Experience

- Undergraduate Teaching Assistant | Brigham Young University | 2022–2023. Assisted students in Computation Theory, Discrete Structures, and Introduction to Computer Systems, helping with proofs, mid-size programming projects, and complex theoretical concepts. Provided accessible explanations to students from diverse backgrounds, including overcoming language and gender barriers in technical disciplines. Received positive feedback for clarity and dedication to student success.
- Graduate Teaching Assistant | Brigham Young University | 2025–Present. Assisted students in Game Theory and Theory of Predictive Modeling. Advanced course in mathematical game theory based on the textbook by Joao P. Hespanha, and coursework development for a mathematical and philosophical course on good modeling techniques.
- Chair of the Operations Research and Engineered Systems lab at Brigham Young University, 2025-Present | Organized meetings, a training/lecture series on related topics, and assisted undergraduates in publishing their own research

Strengths and Achievements

- Critical Analysis | Experienced in identifying and solving complex security problems in cyber-physical systems through robust mathematical frameworks.
- Problem-Solving | Persistent in approaching open-ended challenges, particularly in theoretical and practical system security contexts.
- Academic Excellence | Published research, presented at leading conferences, and earned Cum Laude-level honors in computer science.
- Team Leadership | Effective in managing logistics, well-being, and collaboration for both academic and field projects.

Other Professional Experience

- QA Auditor. Growth LLC | 2021–2023. Audited client calls and summit presentations for a life and business coaching company, providing detailed feedback for quality improvement. Developed strong interpersonal communication skills and gained insights into process improvement and organizational growth.
- Enumerator | National Agricultural Statistics Service (NASDA), Department of Agriculture | 2016–2021. Conducted fieldwork to collect crop data and random samples for agricultural yield estimation, adhering to federal data collection protocols. Navigated logistical challenges and ensured accurate data for government reporting.
- Religious Missionary | The Church of Jesus Christ of Latter-Day Saints | 2017–2019. Led and managed the well-being and logistics of a 14-member team in the Marshall Islands/Kiribati Mission. Gained experience in cultural adaptability, leadership, and coordinating team operations in dynamic environments.